Mathematical Statistics II

Hypothesis Testing

Jesse Wheeler

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1 Introduction

Introduction

- One way to address parameter uncertainty is considering the variance of our estimator, $\hat{\theta}$.
- In some cases, the exact variance can be calculated, using known distributions. In other cases, it needs to be approximated (e.g., observed-information at the MLE).
- Alternative ways of accounting for uncertainty is in the form of *hypothesis tests* and *confidence intervals*.
- This material is largely based on Rice (2007, Chapter 9), Casella and Berger (2024, Chapter 8), as well as some supplemental material from Pawitan (2001).

Rough plan

- Introduction. Comment / reference to Bayesian hypothesis testing (CB 8.2.2)
- Definition (Casella + Berger 8.1.1, 8.1.2, 8.1.3). Follow with silly example, random tests.
- Introduce likelihood ratios (CB 8.2.1)
- Error types and the power function (CB 8.3.1)

- Neyman-Pearson Paradigm (i.e., fix level α before test, Rice 9.2) + Neyman-Pearson Lemma (CB Theorem 8.3.12)
- Relation of NP to Sufficient Statistics (CB Corollary 8.3.13)
- Karlin-Rubin Theorem (composite hypotheses, CB 8.3.17)
- P-values (CB 8.3.26)
- Confidence Intervals (Rice 9.3)
- Profile likelihoods + Wilk's Theorem (Pawitan 3.4)

Definitions

Definition: Hypothesis
A *hypothesis* is a statement about a population parameter.

- Example: The mean height μ of a population is smaller than 6 ft.
- The definition of a hypothesis is quite general. The important part is that it makes a statement about a population parameter.
- The goal of a *hypothesis test* is the decide, based on a sample from the population, which hypotheses are true.
- In practice, we usually perform a hypothesis test by considering two complimentary hypotheses. Our goal is to decide which is true, based on data.

Definition: Null and Alternative Hypothesis
The two complementary hypotheses are called the *null hypothesis* and the *alternative hypothesis*, denoted H_0 and H_1, respectively.

- For population parameters, we usually consider hypotheses of the form:

$$H_0 : \theta \in \Theta_0 \quad \text{and} \quad H_1 : \theta \in \Theta_0^c,$$

  where Θ_0 is some subset of the parameter space.

Blood Pressure medication
Suppose we are interested in testing a new blood pressure medication against existing treatments.
The population parameter of interest is θ, the mean change in blood pressure of this new drug, relative to existing treatments.
We may be interested in testing if $\theta \in \Theta_0 = \{0\}$, against $\theta \in \Theta_0^C = \mathbb{R}/\{0\}$, or:

$$H_0 : \theta = 0, \quad H_1 : \theta \neq 0.$$

- In a *hypothesis testing problem*, the experimenter must decide to either accept H_0 (or equivalently, reject H_1), or accept H_1 (or equivalently, reject H_0).
- We take the approach of Casella and Berger (2024), and not worry about the particular language used, rather focus on the mathematics and final decision.

Definition: Hypothesis tests
A *hypothesis testing procedure*, or *hypothesis test* is a rule that specifies:

(i) For which sample values the decision is made to accept H_0 as true.

(ii) For which sample values H_0 is rejected and H_1 is accepted as true.

The subset of the sample space for which H_0 is rejected is called the *rejection region* or *critical region*. The complement of the rejection region is called the *acceptance region*.

- Rather than specifying individual possibilities for the sample space, we generalize by considering a *test statistic*: $W(X) = W(X_1, X_2, \ldots, X_n)$, which is a function of the sample.

Example: Coin Flipping
Suppose we have a new coin, with probability of heads θ. In our experiment, we will flip a coin $n = 4$ times. Consider making a hypothesis test for $H_0 : \theta = 0.5$ against $H_1 : \theta \neq 0.5$.
First, we can specify our test based on the sample space. The sample space consists of heads and tails:

$$\Omega = \{HHHH, HHHT, HHTH, \ldots, TTTT\}.$$

The null hypothesis states that the probability of tails and heads is equal. Thus, we might define our rejection region as:

$$R = \{HHHH, TTTT\}.$$

That is, we will reject H_0 if we only see heads, or only see tails.
Instead of specifying based on the sample space, we can instead use a sample statistic. Let $X_i = 1$ if the ith coin flip is a heads, and $X_i = 0$ if it is a tails. Then, our sample statistic might be:

$$W(X_1, X_2, X_3, X_4) = \sum_i X_i$$

and we can define the same rejection region using this sample statistic. Namely,

$$R = \{\omega \in \Omega : W(\omega) = 0, 4\},$$

or, equivalently,

$$R = \{\omega \in \Omega : |W(\omega) - 2| \geq 2\}.$$

- In the example above, we could have picked *any* rejection region that we wanted.
- What we would like to do is develop ways to mathematically compare hypothesis testing frameworks.
- We will discuss a few different ways of developing hypothesis tests, and how to compare different testing frameworks.
- Like with point-estimators, this will involve the consideration of some general principles that should be considered.

2 Error types and testing paradigms

Testing Paradigms

- There are various disagreements about how to perform / what to prioritize when conducting hypotheses tests.
- We will categorize these ideals into three main groups:

1. *Bayesian hypothesis testing.*

- The Bayesian treatment of probability as beliefs enables us to consider probabilities that H_0 or H_1 is true, based on the posterior distribution. That is, since $\pi(\theta|x)$ is a density for a random variable $\Theta|X = x$, we can consider $P(H_0 \text{ is true}|X) = P(\theta \in \Theta_0|X)$, and $P(H_1 \text{ is true } |X) = P(\theta \in \Theta_0^C|X)$.
- See Chapter 8.2.2 of Casella and Berger (2024) for more details and examples.

- *Frequentist hypothesis testing*
  - In the Frequentist setting, we cannot consider $P(H_0 \text{ is true}|X)$, since θ is not random. That is, if $\theta \in H_0$, $P(H_0 \text{ is true } |X) = 1$, and zero otherwise. The thing that is random is the data X, not the parameter θ, and consequently the decision to accept or reject is random. Here, we consider the probability of being wrong under various assumptions.

2. *Neyman-Pearson* approach. This has largely dominated hypothesis testing in 20th / 21st centuries. If you've seen a hypothesis test, it's likely based on this approach.
3. *Fisher* approach. Less commonly used, though sometimes mixed with the *Neyman-Pearson* paradigm.

Error Types

- We'll largely focus on the Neyman-Pearson paradigm, since that is the most widely used approach.
- To do so, we need to introduce error types. There are two ways we can make an error:
  1. Conclude that H_0 is false, when it is actually true. This is called a *Type I error*.
  2. Conclude that H_0 is true, when it is actually false. This is called a *Type II error*

Table 1: Decision and error types.

	H_0 True	H_0 False
Accept H_0	correct	Type II
Reject H_0	Type I	correct

- The Neyman-Pearson approach focuses on these two types of error.
- Because of randomness, we can't completely eliminate both (or either) type of error.
- There's also a tradeoff: tests that have larger-rejection regions are more likely to reject, meaning more likely to have a Type I error, but a lower Type II error. Similarly, tests that reject less often are more likely to have Type II errors, less likely to have Type I errors.
- Thus, efforts to decrease one of the error types typically results in an increase of the other type.
- In the frequentist setting, we are interested in long-run frequencies. That is, for any fixed tests, what are the corresponding rates that we make errors.
  - Let α be the Type-I error rate.
  - Let β be the Type-II error rate.
- *Neyman-Pearson Hypothesis Testing*:
  - Control the Type I error rate α at some pre-determined level (e.g., $\alpha = 0.05$). We call this the *significance level* of the test.

- Within this constraint, we want to also minimize β, or equivalently, maximize $1-\beta$, which is known as the *power* of the test.

Example: Flipping a coin
Consider the experiment of flipping a coin $n = 10$ times. We would like to test the hypothesis $H_0 : \theta = 0.5$ against $H_1 : \theta \neq 0.5$, where θ is the probability of heads. Find the rejection region for an $\alpha = 0.05$ level test.
First, recognize from our earlier discussions that we will reject H_0 if the observed proportion of heads $\sum_i X_i/n$ is much different from 0.5. That is, we will reject if we see $\sum_i X_i$ being either very large, or very small. What we need to determine is what is considered "very large" and "very small" in this context.
We would like to control the Type-I error rate at $\alpha = 0.05$; meaning, if we perform this hypothesis many times and H_0 is true, we will be wrong only 5% of the time. Thus, if we assume H_0 is true, we want the probability of our particular sample ω to lie in the rejection region to be smaller than 0.05. We now assume H_0 is true, or that $\theta = 0.5$, and calculate the probabilities of individual samples $X \in \{0, 1, 2, \ldots, 9, 10\}$.

$$\begin{aligned}
P_{\theta\in H_0}(X = 0) &= 0.5^{10} \approx 0.00098 \\
P_{\theta\in H_0}(X = 1) &= 10 \times 0.5^{10} \approx 0.00977 \\
P_{\theta\in H_0}(X = 2) &\approx 0.04395 \\
&\vdots \\
P_{\theta\in H_0}(X = 8) &\approx 0.04395 \\
P_{\theta\in H_0}(X = 9) &= 10 \times 0.5^{10} \approx 0.00977 \\
P_{\theta\in H_0}(X = 10) &= 0.5^{10} \approx 0.00098
\end{aligned}$$

From these probabilities, we can construct our rejection region R. In particular, we want to select R in a way that controls the Type-I error rate at $\alpha = 0.05$, or that $P(X \in R|H_0) \leq 0.05$. The following rejection regions satisfy this condition:

$$R_1 = \{0\},\ R_2 = \{10\},\ R_3 = \{1\},\ R_4 = \{9\},\ R_5 = \{0, 1\},\ R_6 = \{9, 10\},\ R_7 = \{2\},\ R_8 = \{8\},\ R_9 = \{0, 1, 9, 10\}, \ldots$$

among others. Because all of these rejection regions satisfy $P(X \in R_i|H_0) \leq 0.05$. How do we pick which test to use?
Now we recognize that within these conditions, we want to maximize statistical power. That is, what if $\theta \in \Theta_0^C$, or $\theta \neq 0.5$? In this case, we want to increase the probability of rejecting! Thus, we want to pick the region R_i that has the largest probability of rejecting, given that $\theta \neq 0.5$.
This is a harder calculation, since $\theta \neq 0.5$ gives a large number of possibilities. However, we can immediately eliminate regions R_i that are subsets of other regions. That is, if both R_i and R_j control the Type-I error rate at level α, we want to pick one with higher power, and if $R_i \subset R_j$, then:

$$P_\theta(X \in R_i|H_1) \leq P_\theta(X \in R_j|H_1),$$

regardless of the value $\theta \in H_1$.
From this, we can eliminate $R_1, R_2, R_3, R_4, R_5, R_6$ as candidates, since they are all subsets of $R_9 = \{0, 1, 9, 10\}$. This leads us to compare the power of a test based on R_9 against one based on sets like $R_{10} = 2, 10$. One idea is to plot $\beta_9(\theta) = P_\theta(X \in R_9)$ and $\beta_{10}(\theta) = P_\theta(X \in R_{10})$, as a function of θ.
In Figure 1, we see that we get higher-power using the test based on rejection region R_9 whenever $\theta < 0.23$, or $\theta > 0.55$. However, in the interval $\theta \in (0.23, 0.55)$, the test with region R_{10} is more powerful. If there is a strong prior belief that $\theta \in (0.23, 0.55)$, then the test using rejection region R_{10} may be preferred.

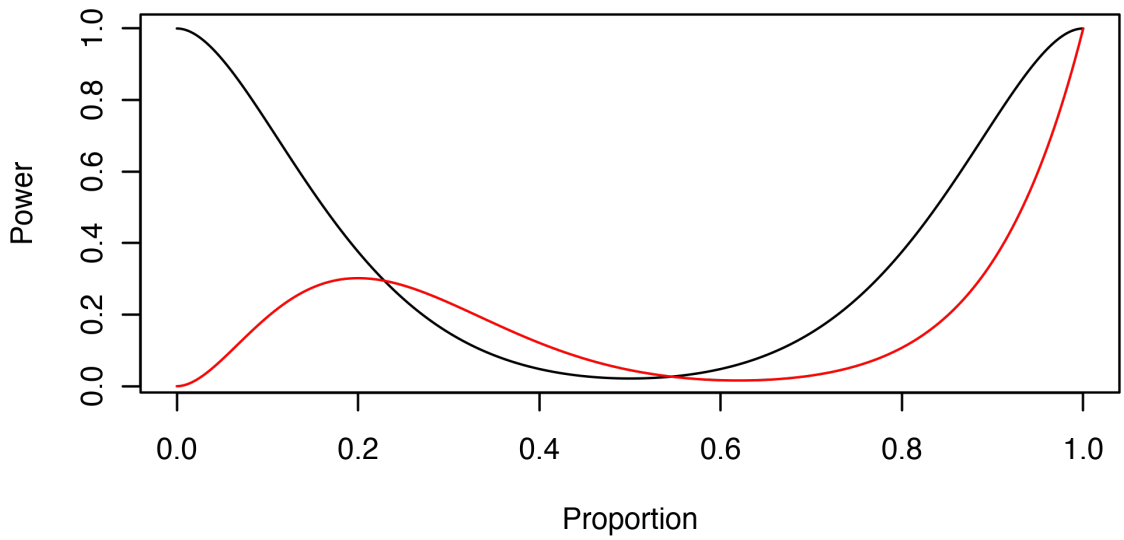

Figure 1: Power functions for two possible hypothesis tests, with corresponding rejection regions R_9 (black line) and R_{10} (red line), respectively.

However, the region R_{10} as a very odd property: when $\theta \in (0, 0.2)$, the power *decreases* the further away we get from the null hypothesis $\theta = 0.5$. We call this type of a test *biased*.

Definition: biased tests
A test with power function $\beta(\theta)$ is *unbiased* if $\beta(\theta_1) \geq \beta(\theta_2)$ for every $\theta_1 \in \Theta_0^C$ and $\theta_2 \in \Theta_0$. That is, the power is at least as large for all values belonging to the alternative hypothesis than the power for values belonging to the null hypothesis.

One way to avoid biased tests is to design rejection regions so that more extreme deviations from the null hypothesis are not excluded if a less-extreme value is included. For example, if $2 \in R$, then $0, 1 \in R$, since these are more extreme deviations from the null.
From this logic, we can conclude that the test based on the rejection region $R_9 = \{0, 1, 9, 10\}$ is the most powerful $\alpha = 0.05$ level, unbiased test.

- What we derived on the previous slide is a type of *exact hypothesis test*. That is, because we knew the exact distribution of both the data $X_1, \ldots, X_n$ and the sample statistic $W(X_1, \ldots, X_n) = \sum_i X_i$, we could pick a rejection region based on the distribution of $W(X_1, \ldots, X_n)$.
- Other type of exact tests are available. Perhaps the most famous is the t-test: If X_i are iid Normal(μ, σ^2), then $T = (\bar{X} - \mu)/(S/\sqrt{n})$ follows *exactly* a t-distribution with $df = n - 1$ (see Chapter 6 from last semester).
- Most often, however, an exact test is not available, and we will need an alternative way to derive tests and rejection regions.
- Before we present these methods, we will introduce P-values and Fisher's approach to hypothesis testing.

P-values and Fisher's approach

- Informally, the P-value can be interpreted as the probability of observing something as extreme or more extreme, if H_0 is true.
- If the P-value is small, that provides evidence that H_0 is not true.
- A technical definition of a P-value is given in 8.3.26 of Casella and Berger (2024).
- The rejection-region framework presented above is mathematically equivalent to computing a P-value, and comparing to the test size $\alpha = 0.05$.
- In the binomial example, consider observing $X = 1$. With respect to $H_0 : \theta = 0.5$ and $H_1 : \theta \neq 0.5$, there are 3 other events that are as extreme (or more extreme) than $X = 1$, namely: $X \in \{0, 1, 9, 10\}$.
- Thus, the P-value is: $P_{\theta=0.5}\big(X \in \{0, 1, 9, 10\}\big) \approx 0.022$.
- Since P-value $< \alpha = 0.05$, we reject the null hypothesis and conclude that H_1 is true.
- The Neyman-Pearson framework used above implies: the long-run error Type I error rate of repeating this test is less than $\alpha = 0.05$, so we have controlled the Type I error rate.
- There are many growing complaints about this framework. Two that have persisted (and are relevant to our discussion) include:
  - The choice $\alpha = 0.05$ (or any other value) is completely arbitrary. What tolerance you have for a Type I error rate should be problem specific.
  - This framework makes the most sense when a study is going to be repeated, since it is formally a statement about the accuracy of many repeated tests, not our single example.

    For instance, consider this famous anecdote used against this framework: A manufacturing company orders parts from a supplier, and states that they would accept $\alpha = 0.01$ proportion of defectives. The supplier than sent them two seperate boxes, a large one marked "perfect", and a smaller box labeled "defectives". The moral: a global confidence statement about long-run repeatability can be meaningless when there are *recognizable* subsets for the statement is either true or false.
- Building off of our example, suppose instead we are testing $H_0 : \theta = 0.5$ against a *one-sided* alternative: $H_1 : \theta < 0.5$, and then we observe $X = 2$. In this case, the P-value is:

$$P = P_{\theta=0.5}\big(X \in \{0, 1, 2\}\big) \approx 0.055 > \alpha = 0.05.$$

- In this example, the Neyman-Pearson framework would select H_0, and not H_1, without any statement about how close this result is to the arbitrarily chosen $\alpha = 0.05$.
- The Fisherian interpretation of hypothesis tests addresses this issue: rather than picking a predetermined α value (which is arbitrary), we consider the P-value as a continuous measure of evidence against H_0, rather than a binary decision.
- This approach isn't without difficulties, as it is more subjective and does not immediately result in a decision in regards to H_0 and H_1.
- Despite there existing many valid arguments why a Bayesian framework or Fisherian interpretation may be preferred, the Neyman-Pearson paradigm has largely "won" in modern science, to the extent that many publication venues will question why you didn't use Neyman-Pearson if you try another approach.

- The American Statistical Association made a formal statement, somewhat discouraging the use of the Neyman-Pearson approach (and P-values in general) in 2016 (Wasserstein and Lazar, 2016). This has been extremely impactful, and has led to much debate and some controversy (Ionides *et al.*, 2017).

3 Likelihood Ratio Tests

4 Evaluating tests

5 Confidence Intervals

5.1 Exact Confidence Intervals

5.2 Wilk's Theorem

5.3 Profile Likelihoods

5.4 Bootstraping

6 Bias reduction methods

Acknowledgments

- Compiled on March 24, 2026 using R version 4.5.1.

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- We acknowledge students and instructors for previous versions of this course / slides.

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